Year 1 AS Statistics and Mechanics

Chapter 4

Correlation

Practice Booklet

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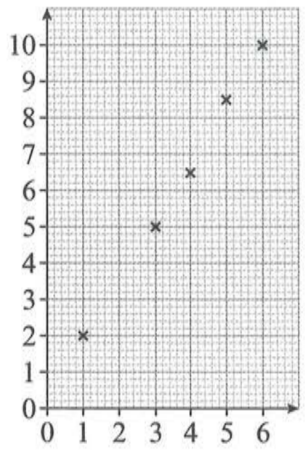
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4.1 Correlation

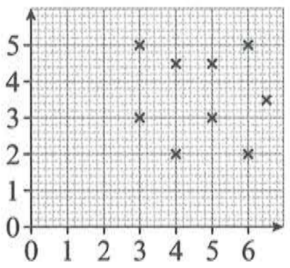
Practice Book, printed page 26.

1. State the type of correlation shown by each of these scatter diagrams.

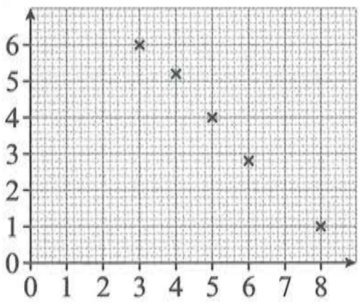
a



b



c



Hint

Look for a pattern in the data.

2. The table shows the results of an experiment.

x	10.1	9.6	8.2	7.4	6.1	5.8	4.1	3.7	3.1	2.5
y	1.2	2.3	3.7	4.5	4.9	4.3	5.9	6.1	6.5	8.1

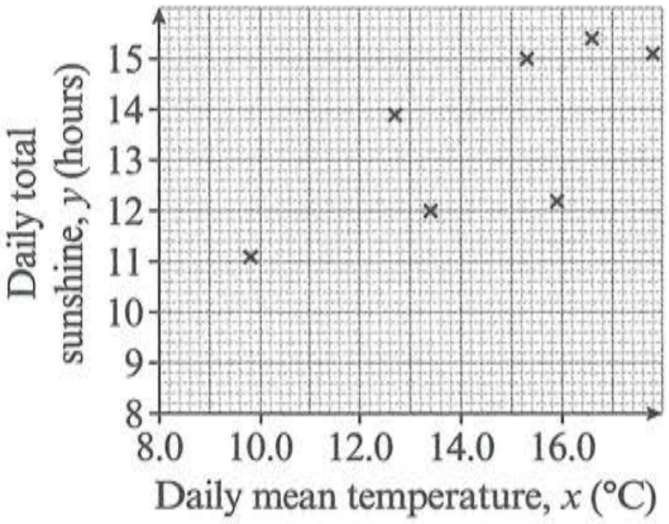
a. Draw a scatter diagram for this data.

b. Describe the type of correlation shown.

Hint

Plot the bivariate data points as coordinates.

3. E The daily mean temperatures, $x\,^{\circ}\text{C}$, and the daily total sunshine, y hours, were recorded on 7 days in one month in 1987 in Camborne.
The scatter diagram shows this data.



a. Describe the type of correlation shown by the scatter diagram. (1 mark)

b. Interpret the correlation in context. (1 mark)

c. Using your knowledge of the large data set, state which month these data might have been sampled from. (1 mark)

4. E/P The engine sizes in litres and the numbers of miles per gallon are recorded for seven cars.

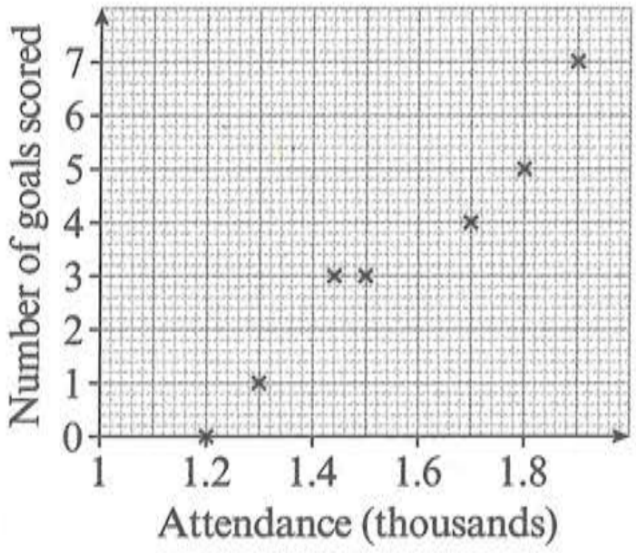
Engine size (litres)	1.1	1.3	1.4	1.6	1.8	2.0	2.4
Miles per gallon	58	52	49	45	42	40	32

a. Draw a scatter diagram for this data. (3 marks)

b. Describe and interpret the correlation between the two variables. (2 marks)

c. State, with a reason, whether it is reasonable to conclude that there is a causal relationship between engine size and mileage. (1 mark)

5. E/P Lenaee is a keen follower of Straston Town Football Club. She collects data on the crowd attendance, in thousands, at seven home games and records the total number of goals scored in each match. She displays this information in a scatter diagram.



a. Describe the correlation shown in the scatter diagram. (1 mark)

Lenaee says that there is a causal relationship between the size of the crowd and the total number of goals scored.

b. Comment on Lenaee's claim. (2 marks)

6. E/P The table shows the daily mean visibilities, v Dm, and the daily mean pressures, p hPa, in Leeming for a random sample of 10 days in July 2015.

Daily mean visibility, v	1100	2800	2900	2800	2600	3300	3500	3200	3300	3600
Daily mean pressure, p	1004	1007	1007	1007	1008	1011	1013	1013	1013	1015

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The median and quartiles for the visibility data are:

$$Q_1 = 2800 \qquad Q_2 = 3050 \qquad Q_3 = 3300$$

An outlier is defined as a value which lies either more than 1.5× the interquartile range above the upper quartile or more than 1.5× the interquartile range below the lower quartile.

a. Show that $v = 1100$ is an outlier. (1 mark)

b. Give a reason why you might:

i. include

ii. exclude this day's readings.

(2 marks)

c. Exclude the day's readings and draw a scatter diagram to represent the data for the remaining nine days. (3 marks)

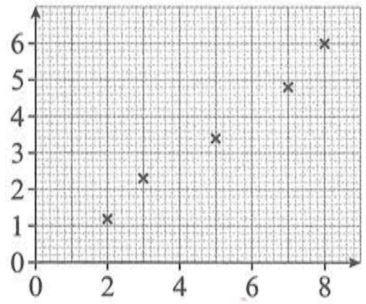
d. Describe the correlation between daily mean visibility and daily mean pressure. (1 mark)

e. Using your knowledge of the large data set, explain whether there could be a causal relationship between the daily mean visibility and the daily mean pressure. (1 mark)

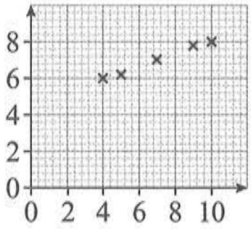
4.2 Linear regression

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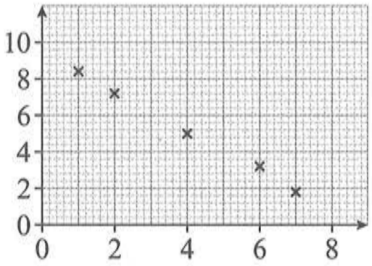
1. Three scatter diagrams are shown below.



Graph A



Graph B



Graph C

Match the regression equations to the graphs.

a. $y = 0.75x - 0.19$

b. $y = 9.30 - 1.05x$

c. $y = 0.35x + 4.60$

Hint

Consider the gradients and y-intercepts of the lines of best fit.

2. It is known that cola can erode a copper coin. A newly minted 1p coin has mass 3.56 g. Hannah decides to immerse the new coin in cola and record its mass, x grams, after t hours. She finds that the equation of the regression line of x on t for her data is $x = 3.56 - 0.012t$. For the regression equation, interpret the meaning of the values:

a. 3.56

b. -0.012

Hint

The regression line shows the trend over time.

3. E/P Sanjeevan collects data on the value, y £1000s, of office blocks in the business district of his local town, and on the number of offices, x, within those office blocks. He calculates the equation of the regression line for y on x to be $y = 50 + 20x$.

a. Use Sanjeevan's regression line to find an estimate for the value of an office block with 12 offices. (1 mark)

b. Interpret the meaning of the number 20, the coefficient of x, in the equation. (1 mark)

Sanjeevan claims that the value of a building in the business district with no offices is £50 000.

c. State, with a reason, whether this is likely to be the case. (2 marks)

4. E/P Scarlett is a zoologist and collects data on the masses, x g, of Siberian chipmunks, t days after they are born. The data is shown in the table.

Days after birth, t	1	2	4	6	7	8	10
Mass of chipmunk, x **(g)**	7.5	12	21	30	34.5	39	48

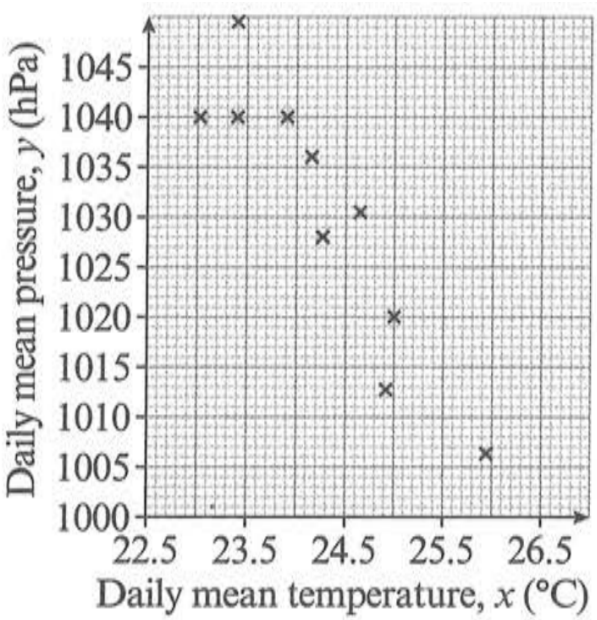
She calculates the equation of the regression line of x on t to be $x = 3 + 4.5t$.

a. Interpret the meaning of the values 3 and 4.5 in the regression equation. (2 marks)

b. Explain why it would not be sensible to use the regression line to predict the mass of a chipmunk that is 30 days old. (1 mark)

c. Explain why it would not be appropriate to use the regression line to predict the number of days after birth for a chipmunk weighing 32 g. (1 mark)

5. E/P The daily mean temperatures, $x\,°\text{C}$, and the daily mean pressures, y hPa, are recorded for a random sample of 10 days in Beijing in June 2015. The data is displayed in a scatter diagram.



The equation of the regression line of y on x is calculated to be $y = 1043 - 1.41x$.

a. Comment on the validity of using the regression line to estimate the daily mean pressure when the daily mean temperature is:

i. 24.5°C

ii. 28.2°C

(2 marks)

b. Explain why it would not be appropriate to use the regression line to predict the daily mean temperature when the daily mean pressure is 1009 hPa. (1 mark)

c. With reference to correlation, comment on how accurately the regression model reflects the data. (2 marks)

Problem solving

Bronze

Some meteorologists are interested in the relationship between the daily minimum temperature, $x\,°\text{C}$, and the daily maximum temperature, $y\,°\text{C}$, in a particular town. They take a random sample of 7 days and record the data in a table.

Minimum temperature, x (°C)	0.6	1.1	1.7	2.8	3.3	5.0	7.2
Maximum temperature, y (°C)	12.2	13.9	14.4	16.7	17.2	19.4	22.2

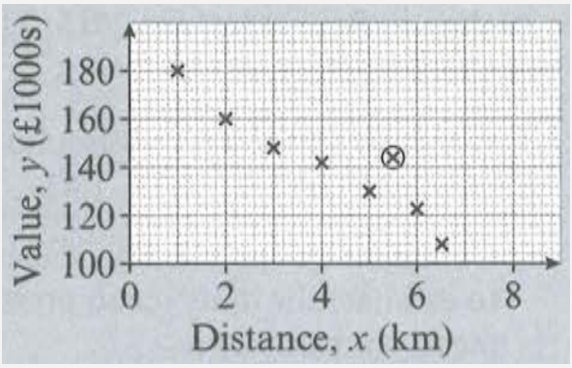
a. Draw a scatter diagram to represent this data. (2 marks)

b. Explain why a linear regression model may be appropriate to describe the relationship between the daily minimum temperature and the daily maximum temperature. (1 mark)

c. Suggest, with a reason, whether it would be sensible to use the regression model to predict the daily maximum temperature when the daily minimum temperature is 12°C. (1 mark)

Silver

A survey of 10 two-bedroom flats in a particular town is carried out. The value of the flats, y £1000s, and the distance, x km, from the town centre is recorded.
A scatter diagram is drawn to show the data.



a. Give a reason to support the use of linear regression to model this data. (1 mark)

The equation of the regression line of y on x is found to be $y = 185 - 10x$.

b. Interpret the values 185 and -10 in the regression equation. (2 marks)

c. Explain why it would not be sensible to use the regression model to find y when $x = 11$. (1 mark)

The point representing one of the flats has been circled on the graph.

d. State, with a reason, whether you think this point is likely to represent a recording error or a valid data value. (1 mark)

Gold

A researcher is investigating the link between humidity and visibility using the large data set. The researcher analyses the data for Heathrow in 2015, and considers the daily maximum relative humidity, x%, and the daily mean visibility, y 100 Dm, for each day.
The equation of the regression line for y on x is found to be $y = 113 - 0.933x$.

a. Give an interpretation of the value -0.933 in the regression equation. (1 mark)

b. Using your knowledge of the large data set, suggest, with a reason, whether it would be sensible to use this regression model to estimate the mean visibility on a day when the daily maximum relative humidity is:

i. 90%

ii. 40%

(2 marks)

c. Using your knowledge of the large data set, explain whether there could be a causal relationship between these variables. (1 mark)

d. Explain why it would not be appropriate to use the regression line to find the humidity on a day when the visibility is 3500 Dm. (1 mark)

Now try this → Exam question bank Q4, Q8, Q12, Q19, Q21

Selected exam question bank questions

Question 4

Shania conducted a scientific experiment to measure the temperatures, $h\,°\text{C}$, generated by a chemical reaction and the amounts of a catalyst, x g, that was used in the reaction.

The smallest amount of catalyst she used was 1 g and the largest amount of catalyst she used was 30 g.

She calculated the regression line of h on x for her data and found it to be $h = 30 + 2.1x$.

a. Interpret the figures 30 and 2.1 in the context of the question. (2 marks)

Shania decides to use her regression line to predict the temperature of the reaction if she uses 100 g of the catalyst.

b. Explain why it would not be sensible to do this. (1 mark)

Her lab partner Monique uses the regression line to predict the amount of catalyst used when the temperature generated by the reaction is 60°C.

c. Explain why it is not appropriate for Monique to do this. (1 mark)

Question 8

The table shows the ages, t years, and the second-hand sale prices, y £1000s, of seven sports cars of the same model and specification at a particular dealership in one month.

Age, t years	0.1	0.5	0.8	1.1	1.4	1.5	1.7
Value, y £1000s	120	118	112	100	30	71	35

Elsbeth thinks a linear regression model is suitable for modelling this data. She calculates the equation of the regression line for y on t to be $y = 120 + 0.1t$.

a. Explain how you know that Elsbeth has made a mistake with her regression line calculation. (1 mark)

b. Assess Elsbeth's claim that a linear regression model is suitable for modelling this data. You must provide evidence for your conclusion. (2 marks)

One of the data points does not fit the general trend of the data.

c. Suggest a possible reason why this might still be a valid data point. (1 mark)

Question 12

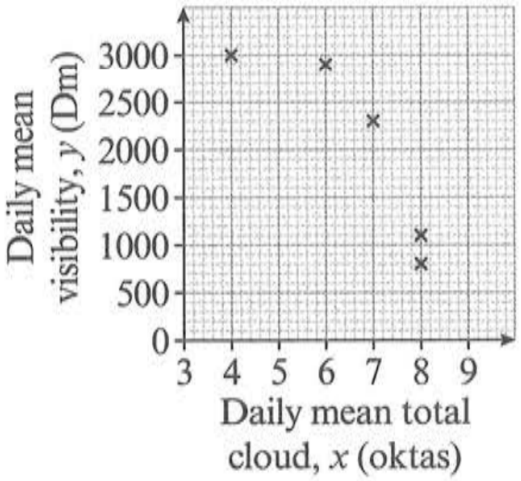
Andrea is investigating the relationship between cloud cover and visibility. She takes a random sample of five days from the large data set for Hurn in May 1987. The table records her data.

Daily mean total cloud, x (oktas)	8	4	6	7	8
Daily mean visibility, y (Dm)	800	3000	2900	2300	1100

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a. Using your knowledge of the large data set, explain what is meant by an 'okta'. State whether this variable is discrete or continuous. (2 marks)

Andrea draws a scatter diagram of this data.



She then calculates the equation of the regression line for y on x as $y = 5530 - 532x$.

b. Suggest a reason why Andrea should not have used a linear regression model for this data. (1 mark)

c. Suggest two more criticisms of Andrea's method. (2 marks)

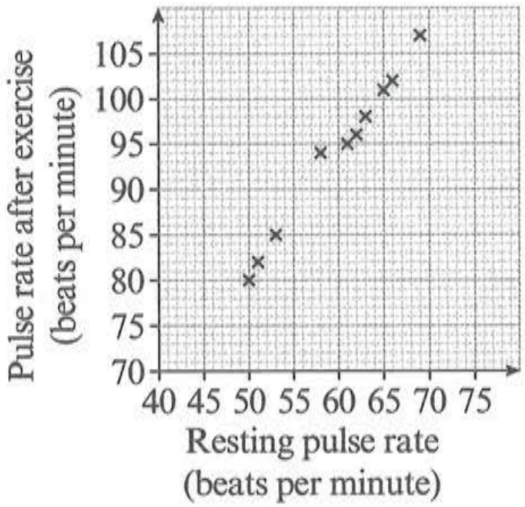
Question 19

Adrian is doing an experiment to find out whether there is a link between the pulse rate of a person after exercise and that individual's resting pulse rate.

He asks the first 10 students that he sees to take part in the experiment. He measures each student's resting pulse rate, x beats per minute, and also their pulse rate after exercise, y beats per minute.

a. State the sampling method that Adrian has used. (1 mark)

The scatter diagram shows the results of Adrian's experiment.



b. Describe the correlation between the pulse rate after exercise and the resting pulse rate. (1 mark)

Adrian calculates the equation of the regression line for y on x as $y = 13.0 + 1.36x$.

c. Explain why Adrian is justified in using a linear regression model for this data. (1 mark)

d. Interpret the figure 1.36 in the context of the question. (1 mark)

e. Explain why it would not be sensible for Adrian to use his equation to calculate the pulse rate after exercise for an 11th student whose resting pulse rate is 45 beats per minute. (1 mark)

Question 21

The table shows the daily mean windspeed, w knots, and the daily maximum gust, g knots in Leeming for a random sample of 9 days in 2015, taken from the large data set.

Windspeed, w (kn)	5	7	7	5	8	6	6	7	7
Max gust, g (kn)	18	19	19	18	31	18	17	21	19

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The median and quartiles for the maximum gust data are:

$$Q_1 = 18 \qquad Q_2 = 19 \qquad Q_3 = 20$$

An outlier is defined as a value which lies more than 1.5× the interquartile range above the upper quartile or more than 1.5× the interquartile range below the lower quartile.

a. Show that $g = 31$ is an outlier. (1 mark)

b. Give a reason why you might exclude this day's readings. (1 mark)

The data is cleaned and the regression line for g on w calculated for the remaining 8 data points. The equation of the regression line is found to be $g = 13.2 + 0.864w$.

c. Interpret the meaning of the figures 13.2 and 0.864 in the context of the question. (2 marks)

d. Do you think there is a causal relationship between the daily mean windspeed and the daily maximum gust? Explain your reasoning. (1 mark)

Answers

4.1 Correlation

1.

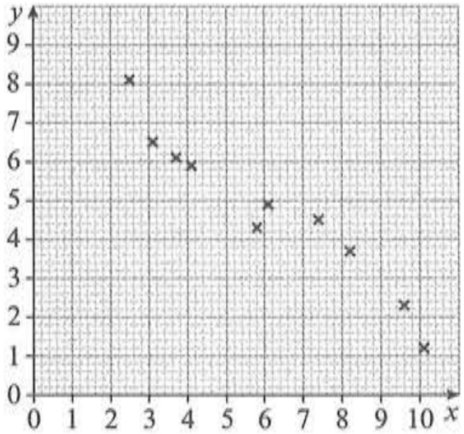
a. Positive correlation

b. No/zero correlation

c. Negative correlation

2.

a.



b. Negative correlation

3.

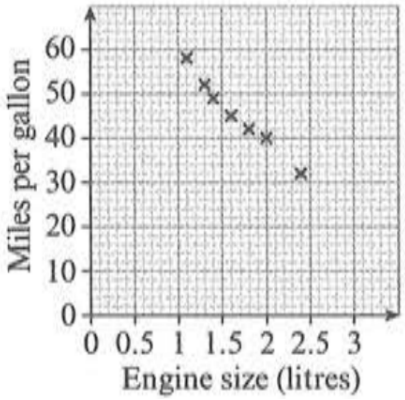
a. Positive correlation

b. As the daily mean temperature increases, the total daily sunshine increases.

c. Accept July, August or September.

4.

a.



b. Negative correlation. As the engine size increases, the car does fewer miles per gallon.

c. Larger engines will need more fuel to run, so reasonable to assume a causal relationship.

5.

a. Positive correlation

b. Unlikely to be causal as you don't know before the game how many goals will be scored. There could be a third variable that influences crowd size and number of goals scored, such as recent team performance.

6.

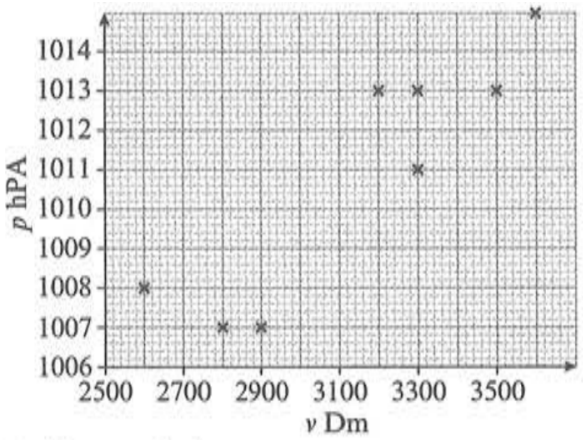
a. $2800 - 750 = 2050$, so 1100 is an outlier.

b.

i. Could have just been a particularly foggy day.

ii. It is a long way away from the rest of the data so could have been recorded in error.

c.



d. Positive correlation.

e. Yes. High pressure can cause less rain/cloud so could improve visibility.

4.2 Linear regression

1.

a. Graph A

b. Graph C

c. Graph B

2.

a. The initial mass of the coin is 3.56 g.

b. The mass of the coin decreases by 0.012 g every hour.

3.

a. £290 000

b. The increase in value (in £1000s) of an office block for each additional office.

c. No. Buildings with no offices are likely to be retail outlets, or residential properties, and therefore will not fit the pattern of his data concerning office blocks.

4.

a. The mass, in grams, at birth; the increase in mass each day.

b. This is outside the range of the data.

c. This is the regression line for x on t. To predict age from mass, you would need the equation of the regression line for t on x.

5.

a.

i. Valid: inside the range of the data.

ii. Invalid: outside the range of the data (extrapolating).

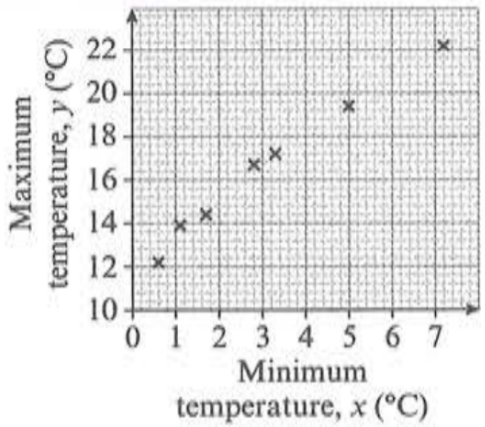
b. This is the regression line for y on x. To predict temperature from pressure you would need the regression line for x on y.

c. The data shows some degree of linear correlation, which suggests that the linear regression model will be a reasonable model for the data.

Problem solving

Bronze

a.



b. The points appear to be close to a straight line.

c. No: 12 is outside the range of the data, so this would be extrapolation.

Silver

a. The data exhibits a (very strong) linear trend.

b. The value, in £1000s, when the flat is right in the centre of the city; the decrease in value for every 1 km from the city centre.

c. Outside the range of the data, so this would be extrapolation.

d. It is likely to be valid, as the value is not unrealistic. It could be that that flat is in a more expensive area than the others sampled, or it could have significantly bigger rooms.

Gold

a. The visibility decreases by 93.3 Dm for each 1% increase in humidity.

b.

i. 90% is within the range of the data, so would be sensible.

ii. The daily maximum relative humidity in the UK never falls below about 60%, so not valid.

c. High humidity is associated with mist and fog, so it is likely that the relationship could be causal.

d. This is the regression line for y on x. To predict the humidity from the visibility you would need the equation of the regression line for x on y.

Selected exam question bank answers

Question 4

a. 30: the temperature of the reaction with zero catalyst; 2.1: the increase in temperature for each additional gram of catalyst.

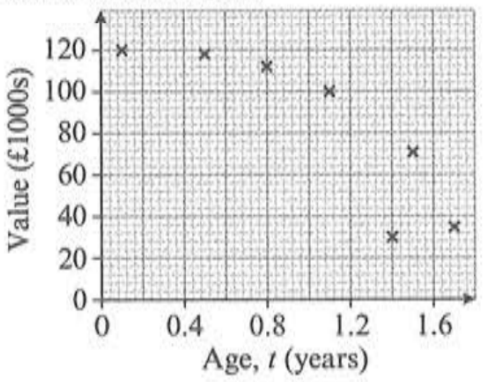
b. 100 is significantly outside the range of data used to calculate the regression equation.

c. This is the regression line for h on x. To estimate the amount of catalyst that generates a specific temperature she should use the regression line for x on h.

Question 8

a. The value goes down as age goes up so the gradient of the regression line should be negative.

b. Evidence will probably be a scatter diagram, but the data does not follow a linear trend so a linear regression model is not suitable.



c. It could be due to very high mileage, or there could be some damage on the vehicle.

Question 12

a. An okta is an eighth (of the sky covered by cloud), and takes integer values from 0 to 8. It is a discrete variable.

b. The points appear to form a curve, rather than a line.

c. Her data set is very small and taken from a single location in a single month.

Question 19

a. Opportunity sampling

b. Positive correlation

c. The points lie in an approximate straight line.

d. The additional bpm after exercise for each bpm for the resting heart rate.

e. It is outside the range of the data (extrapolation).

Question 21

a. $20 + 3 = 23$, so 31 is an outlier.

b. It was recorded in error, or the data was misread.

c. 13.2: the daily maximum gust when the daily mean windspeed is zero; 0.864: the increase in daily maximum gust for each knot the daily mean windspeed increases.

d. Yes: days with higher mean windspeeds will tend to have higher maximum gusts.